

CSCI3170

Introduction to Database Systems

0-outline-2627R1

Slides: Michael Yu

Need to Know - Staff

- Course Instructor
 - Yu, Michael Ruisi* (mryu@cse.cuhk.edu.hk)
 - *Office Location: SHB 127
 - **If want to talk, easiest to catch me after the lecture**
- Tas (SID @link.cuhk.edu.hk OR CSEname @cse.cuhk.edu.hk)
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 - Default first point of contact, reaches all course personnel
 - Issues with assignment, absence, etc
 - Include your student ID
 - Kindly cc course email in your replies
 - Kindly refrain from sending emails between 12am and 6am

On Course Teaching Hours

- Resources distributed through Blackboard
 - Lecture notes, Assignments & Announcements
 - No Lecture Recordings
- **There will be no tutorials in Week one**

Class	Time	Location
Lecture	Mo 5:30PM - 6:15PM	William M W Mong Eng Bldg LT
Lecture	We 4:30PM - 6:15PM	Mong Man Wai Bldg LT1
Interactive Tutorial T01	Tu 2:30PM - 3:15PM	Lady Shaw Bldg LT6
Interactive Tutorial T02	Th 3:30PM - 4:15PM	Y.C. Liang Hall 103

Tentative Schedule

- *Subject to change in response to course pacing; if so will mainly prioritise contents leading up to Transaction Management.*

Week	Topic
1	Introduction/ Entity-Relationship Model
2	Relational Model
3	Relational Algebra
4	SQL
5	SQL
6	Relational Database Design
7	Mid Term Quiz (Location TBC)
8	Relational Database Design
9	Storage
10	Transaction Management
11	Index and Hashing
12	Query Processing
13	Query Optimization

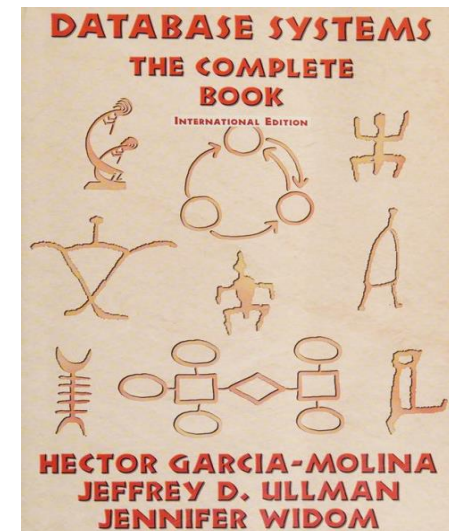
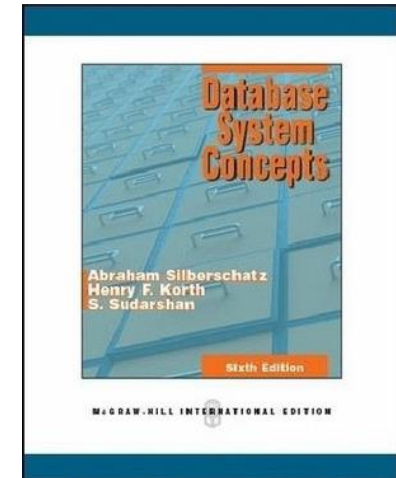
On Assessments

- **Approach 1 – Prohibit all use of AI tools**
 - In any kind of learning activity or assessment that will be counted towards the final course grades (or used for evaluating attainment of the desired learning outcomes), students are not allowed to submit work which is produced with the collaboration of or supported by the use of any AI tools.
- Assessments are all individual work, and...
 - Mid-term will cover content in first half of the course, final will include contents from all the course
 - **Exam hurdle** reinstated: a minimum 30% on final exam itself, no similar hurdle for the mid-term
 - Cheat sheet arrangement is one double-sided A4 paper (Mid-term and final; printed and or written)

Item	Weight (%)	Remarks
Assignments	30	Expect 3, at most 4
Mid Term	20	Week7 Wed (21 st Oct); Location TBA
Final	50	

References

- Database System Concepts
Abraham Silberschatz, Henry F. Korth,
S. Sudarshan., McGraw-Hill, 5th Edition
 - The textbook website is
<http://www.db-book.com>
 - The website includes slides,
solutions to practice exercises.
 - Try practice exercises by yourself!
- Database Systems :
The Complete Book
Hector Garcia-Molina,
Jeffrey D. Ullman, Jennifer Widom
- Database Management Systems
Raghu Ramakrishnan,
Johannes Gehrke, McGraw Hill



Core Motivation and Necessity

- Data will always have to be:
 - **Stored***
 - **Manipulated and or Accessed***
 - **Shared***
 - **Transmitted^**
- Where there is data, there are always **DataBase Management Systems (DBMS)**.
 - Railway and Airline Reservation System. ...
 - Library Management System. ...
 - Banking. ...



Note:

** handled by databases*

^ handled by networks

Top Database Management Systems

- <https://db-engines.com/en/ranking> (yearly updates)
 - Oracle (**Relational** DBMS)
 - MySQL (**Relational** DBMS)
 - Microsoft SQL Server (**Relational** DBMS)
 - PostgreSQL (**Relational** DBMS)
 - MongoDB (Document store)
 - Snowflake (**Relational** DBMS)
 - Redis (Key-value Store)
 - Elasticsearch (Search Engine)
 - IBM DB2 (**Relational** DBMS)
 - SQLite (**Relational** DBMS)

424 systems in ranking, September 2025

Rank			DBMS	Database Model	Score		
Sep 2025	Aug 2025	Sep 2024			Sep 2025	Aug 2025	Sep 2024
1.	1.	1.	Oracle	Relational, Multi-model 	1170.62	-50.08	-115.97
2.	2.	2.	MySQL	Relational, Multi-model 	891.77	-23.69	-137.72
3.	3.	3.	Microsoft SQL Server	Relational, Multi-model 	717.32	-36.84	-90.45
4.	4.	4.	PostgreSQL	Relational, Multi-model 	657.17	-14.08	+12.81
5.	5.	5.	MongoDB 	Document, Multi-model 	380.50	-15.08	-29.74
6.	6.	↑7.	Snowflake	Relational	190.19	+11.29	+56.47
7.	7.	↓6.	Redis	Key-value, Multi-model 	145.17	-2.02	-4.25
8.	8.	↑9.	IBM Db2	Relational, Multi-model 	124.19	-3.12	+1.14
9.	9.	↑14.	Databricks	Multi-model 	124.06	+8.25	+39.82
10.	10.	↓8.	Elasticsearch	Multi-model 	118.26	+3.99	-10.53

- Oracle is available for experimentation in CSCI3170

The Relation (MVP of Relational Theory)

- Database implementations are almost always based on relational model.
 - Extremely simple and useful, the model on which major commercial DBMS's depend today.
 - Uses a simple modelling concept: the *relation*
 - A two-dimensional **table** in which data is arranged
 - We are all very proficient familiar with the general appearance
- The relational model uses a **collection of tables** to represent both the data and the relationships among those data
 - Each table has multiple columns, and each column has a unique name within the table.

sid	sname	asg1	asg2	asg3	exam
1155100001	Rick	90	100	95	88
1155100002	Ryan	100	80	85	76
1155100003	Bruno	100	85	75	82
1155100004	Alice	70	55	80	64
1155100005	Bob	55	30	35	42

In-lecture Scenario

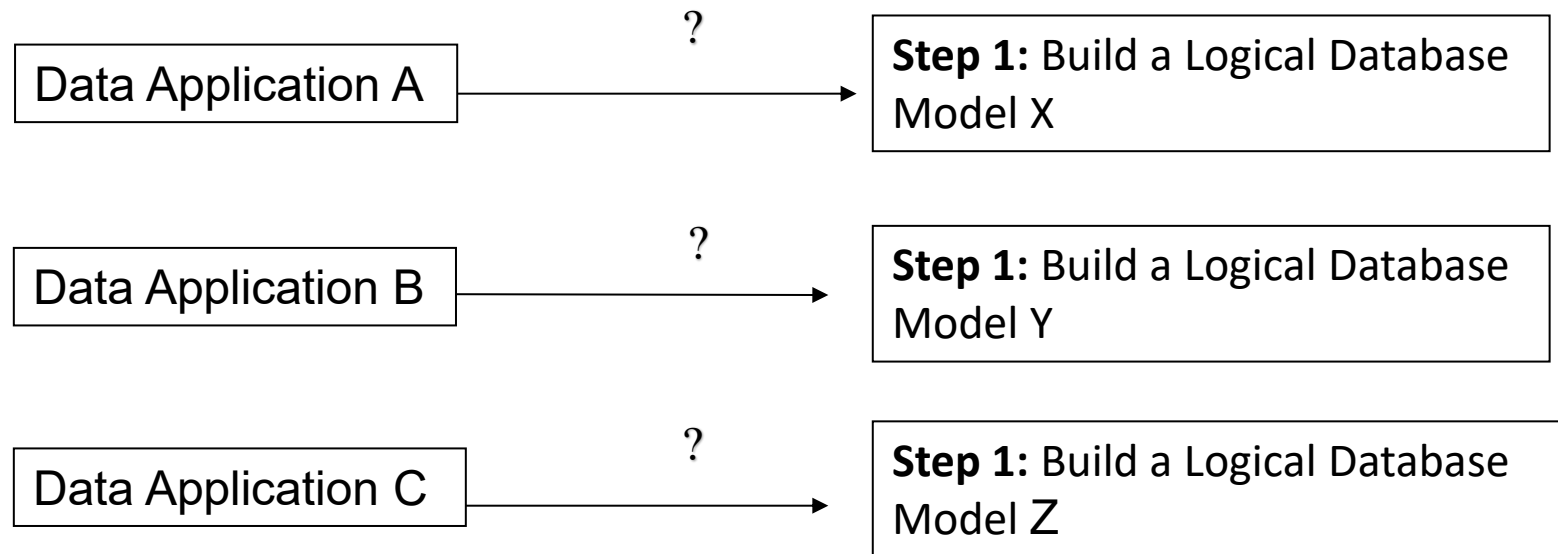
- If we're so familiar; Let's try model!
- CUHK needs us to implement a logical data model/ relational schema storing information about CUHK Students.
 - Let's build a relation to store properties.
 - What columns/ fields are you adding
 - TIP: Process of designing a database begins with an **analysis** of what information the database must hold.

Constraints

- consider case where each field be atomic

A high level database dev. workflow

- Designing a usable logical database schema is generally not too difficult...
 - For a long time, there wasn't a systematic way to do so
 - What is the understanding of good?



- More generally, how should everyone build his/her databases for an application for its given requirements?

Solution Proposed and Accepted!

- Dr. Peter Chen
<https://www.csc.lsu.edu/~chen/>
- His work started a new field of research and practice: Conceptual Modeling
- See paper on Entity-Relationship Model in link
<https://bit.csc.lsu.edu/~chen/pdf/english.pdf>
- Traditionally, database designers begin by developing a conceptual design using the **E/R model**, then translate that into the relational model for implementation
 - Most common model for abstract representation of the structure of a database is the entity-relationship model



Step 1: Do Conceptual Modeling!



Step 2*: Take Conceptual Model and Build the Logic Data Model

(Chen. 1976) E/R Modeling

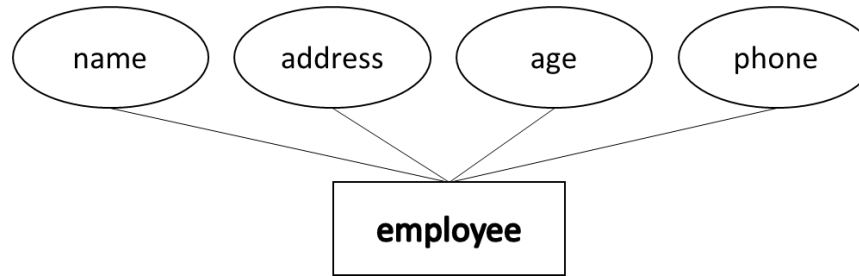
Framework: Entity Relationship Model

- **Elements** of the E/R Model: The structure of data is represented using three principal element types
 - **Entity**,
 - **Relationships**, and
 - **Attributes**.
- In E/R Models, structures are represented **graphically**.
 - Entity sets are represented by **rectangles**
 - Attributes are represented by **ovals**
 - Relationships are represented by **diamonds**

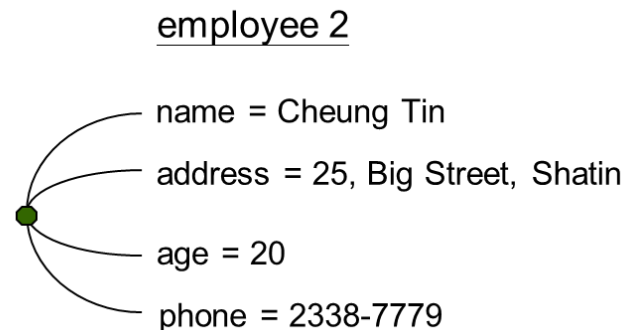
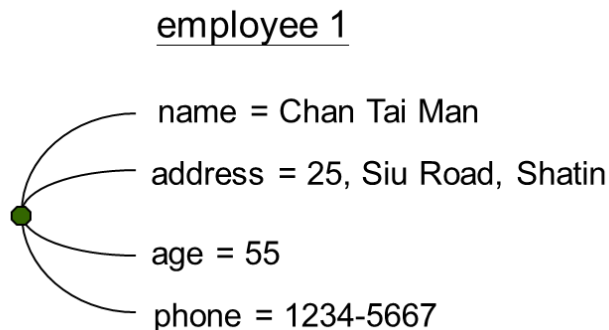
Example

While an *entity type* is the structural blueprint or schema definition, an *entity set* refers to the actual collection of all individual entities belonging to that type *at a given moment*.

- Suppose we want to store info about employees.
- We have an represent an employee **entity type** with the E/R model.
 - Each employee entity defines a fixed number of fields (4).



- Initially, there are zero employees, so the employee **entity set** is empty
 - There are two employee **entity** additions: *Chan* and *Cheung*.
 - Together, they make up the *employee entity set* currently (size of two).

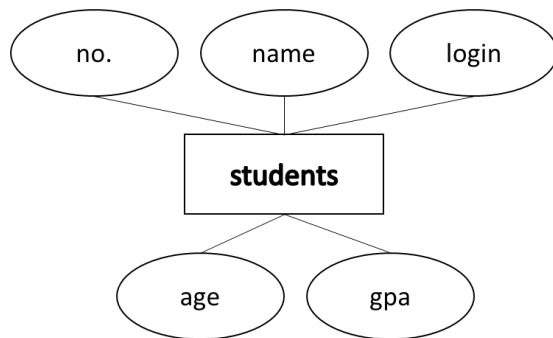


Checkpoint: Terminology

- An **entity** (instance) is a specific object that *exists* and is *distinguishable* from other objects.
 - Example: specific person, company, event, plant
- An **entity type** defines a “class” of objects, with a shared set of specific attributes (corresponding to the table schema)
 - Each entity type provides a description of an entity using a set of **attributes**/ properties.
 - Example:
 - Instructor (ID, name, salary)
 - Course (course_id, title, credits)
- An **entity set** is the group of all entities that share the same type (corresponds to the **entire table** populated with rows).
 - Example: set of all persons, companies, trees, holidays

A Sneak Peak: ER to Rel.

- Let's see what can already be achieved using entities and atomic attributes.
 - Not all concepts are covered yet...
- The relational model is a record-based model
- Each entity type maps to a corresponding relation in the logical model, following Dr. Peter Chen's translation approach.
 - Some additional considerations and exceptions will be discussed later.



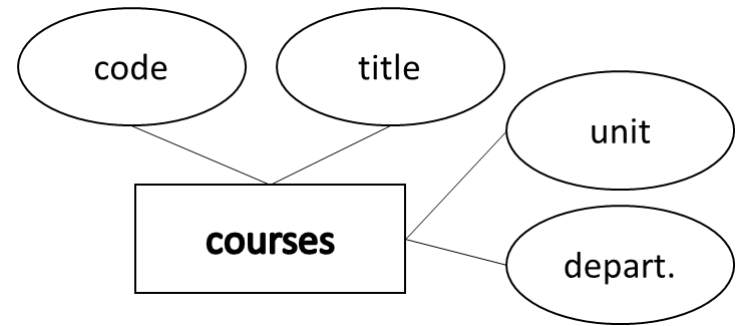
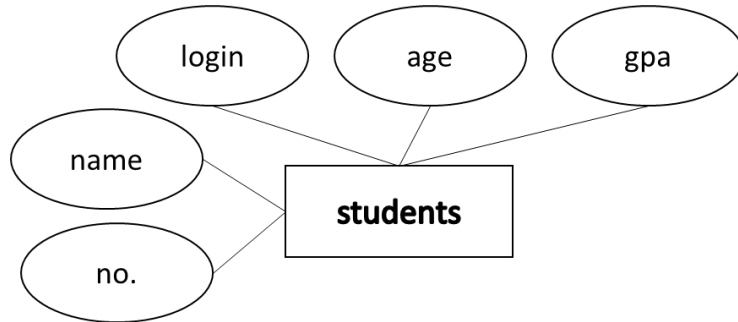
E/R model: students

no.	name	login	age	gpa
53666	Jones	Jones@cs	18	3.4
53688	Smith	Smith@ee	18	3.2
53650	Smith	Smith@math	19	3.8
53831	Madayan	Madayan@music	11	1.8
53832	Guldu	guldu@music	12	2.0

Relational model: students

Isolated Entities? Related information?

E/R POV



students

no.	name	login	age	gpa
53666	Jones	Jones@cs	18	3.4
53688	Smith	Smith@ee	18	3.2
53650	Smith	Smith@math	19	3.8
53831	Madayan	Madayan@music	11	1.8
53832	Guldu	guldu@music	12	2.0

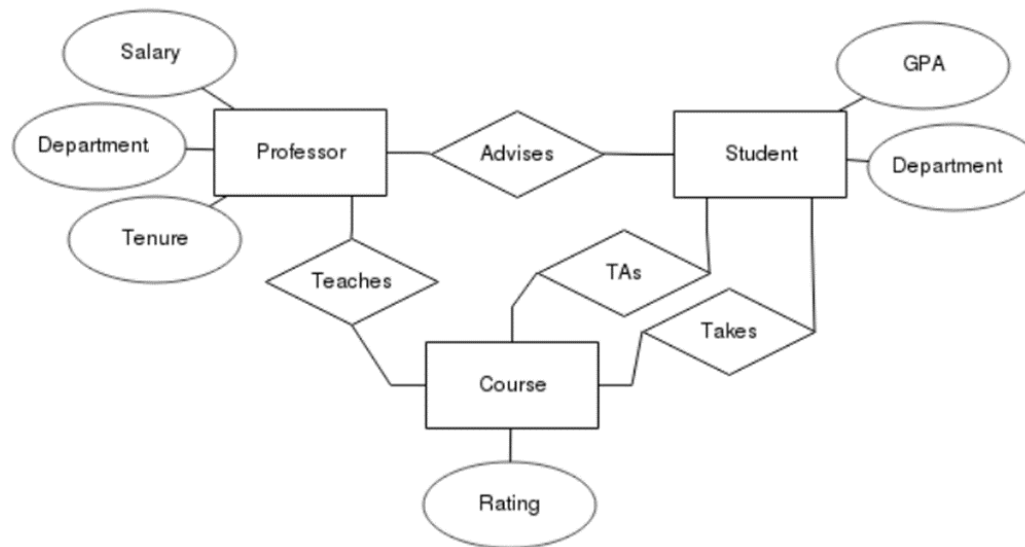
courses

code	title	unit	depart.
CSCI1130	Introduction to Computing Using Java	3	Computer Science
CSCI2100	Data Structures	3	Computer Science
CSCI3170	Introduction to Database Systems	3	Computer Science
ENGG1003	Engineering Design and Communication	3	Engineering
MATH1010	University Mathematics	3	Mathematics

Relation POV

Next: Modelling Relationships

- Represent logical links between two or more entities (or the entity with itself)
 - Entity A may have a relationship with another Entity B
 - Entity A may have a relationship with itself
 - A relationship can also involve Entities A, B, C, ... etc., if need be.
 - More generally, any set of entities are not limited to participating in at most one relationship with each other.



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Final Notes

- **Keywords:**
 - Entity Type (depending on context)
 - Entity (usually an individual object/ instant)
 - Entity Set
 - Attributes (so far just atomic attributes)
 - Relations, and Relational Database
 - E/R Model
- **Key points:**
 - An entity is any object that *exists* and is ***distinguishable*** from other objects (both in E/R, and relational modeling later)
 - Oracle is not the only relational database management system
- In-lecture scenario:
 - Hopefully demonstrate that databases designs are necessary
 - Inspire the exploration of what is a good database schema
- Assumptions:
 - Each entity appears only once in the entity set
 - Each attribute is atomic in value

Acknowledgement

- Some text are inspired from the reference books
 - *Database Systems: The Complete Book by Hector Garcia-Molina, Jeffrey D. Ullman, Jennifer Widom*

and that of

- *Database System Concepts by Abraham Silberschatz, Henry F. Korth, S. Sudarshan., McGraw-Hill, 6th Edition*

Other DB Models support other data.

- Unstructured and Semi-Structured data also present.
- Our course focus is strictly on relational databases, and therefore (structured) “*relational*” data.

Unstructured Data

The university has 5600 students. Shaun (ID Number: 160801), 18 years old Communication study. Linh with ID number 160802, majoring in Accounting and is 20 years old. Ahmed from Psychology study program, 19 years old, ID number 160803.

Semi-Structured Data

```
<University>
<ID Number="160801">
  <Name="Shaun">
  <Age="18">
  <Program="Communication">
<ID Number="160802">
  <Name="Linh">
  <Age="20">
  <Program="Accounting">
..... </University>
```

Structured Data

ID	Name	Age	Program
160801	Shaun	18	Communication
160802	Linh	20	Accounting
160803	Ahmed	19	Psychology

GLEEMATIC A.I.

Appendix

Think about it: not enough attributes?

- Suppose you have two genuinely distinct entities of the same type, but they have the same values.
 - i.e., two distinct entities that need storing but sharing identical attribute values

Physical Object	Brand	Volume	Colour
Coke Can	Coca-Cola	330ml	Red
Coke Can	Coca-Cola	330ml	Red

- Problem:
 - In Entity-Relationship (ER) model, an entity set is defined mathematically as a set; by definition, a set can only contain unique elements and cannot hold duplicate records
 - in the theoretical foundation of a relational database, a relation is also set of tuples.
 - Else entities are no longer functionally distinct
- An **entity** (instance) is a specific object that *exists* and is *distinguishable* from other objects.
 - Challenge, to keep to E/R modelling, we can only use the properties that are naturally available (more on this later)